CBC Digital Learning Platform

Stage G — UI/UX Design

Google Stitch Prompt Compendium

Design System · Screen Prompts · AI Tutor UX · Component Library · Frontend Architecture

Version 1.0 | April 2026 | Draft — Design Phase

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Stage A ✓ Stage B ✓ Stage C ✓ Stage D ✓ Stage E ✓ Stage F ✓
Stage G: UI/UX Design ← Current

Part 0 — Cross-Stage Design Synthesis

Before any screen is designed, this section consolidates every design-relevant requirement across Stages A through F into a single reference that informs every Google Stitch prompt in this document. All prompts in Parts 1–10 derive from these foundations.

0.1 Platform Identity & Design Direction

The CBC Digital Learning Platform is Uganda's national secondary education technology platform. It is not a generic LMS. Its identity must communicate institutional authority, national ownership, and youth-oriented accessibility simultaneously.

Recommended visual identity:

- Primary colour — Deep Institutional Blue: #1B4F72 (trust, authority, academia)
- Secondary colour — Uganda Green: #1E8449 (growth, national identity, competency attainment)
- Accent colour — Warm Amber: #D4A017 (energy, youth, highlights, CTAs)
- Surface — Off-white #F8F9FA with mid-grey #ECF0F1 for cards and panels
- Danger/Error — #C0392B. Success — #27AE60. Warning — #F39C12
- AI Tutor brand colour — Deep Teal #0E6655 (distinct from the blue/green platform palette, signals AI context)

Typography:

- Display / Headings — "DM Serif Display" (Google Fonts) — authoritative, editorial, distinctive
- Body / UI — "Plus Jakarta Sans" (Google Fonts) — modern, legible, friendly at small sizes
- Code / Mono — "JetBrains Mono" — for any AI output, code snippets in lessons
- Scale: 12 / 14 / 16 / 18 / 24 / 32 / 40 / 56px — 8px base grid throughout

0.2 Screen Inventory (All Roles)

Screen ID	Screen Name	Role	SRS Source	Priority
S-AUTH-01	Registration	Learner / Teacher	FR-L-01, FR-T-01	P1
S-AUTH-02	Login	All roles	FR-L-02, FR-T-01, FR-M-01	P1
S-LEARN-01	Learner Home / Feed	Learner	FR-L-03, FR-L-07	P1
S-LEARN-02	Subject & Level Browser	Learner	FR-L-03	P1
S-LEARN-03	Lesson Reader	Learner	FR-L-03, FR-L-04	P1
S-LEARN-04	Offline Library	Learner	FR-L-04, NFR-O-01	P1
S-LEARN-05	Progress Dashboard	Learner	FR-L-09	P1
S-POST-01	Post Feed (Knowledge Wall)	Learner	FR-L-06, FR-L-07	P1
S-POST-02	Create Post	Learner	FR-L-06, FR-L-10	P1
S-POST-03	Post Detail & Comments	Learner	FR-L-07	P1
S-COLLAB-01	Collaborative Sessions	Learner	FR-L-08	P2
S-AI-01	AI Tutor Dedicated Page	Learner	FR-L-05, FR-AI-01	P1

Screen ID	Screen Name	Role	SRS Source	Priority
S-TEACH-01	Teacher Home / Materials	Teacher	FR-T-02, FR-T-03	P1
S-TEACH-02	Upload Material	Teacher	FR-T-02, FR-T-03	P1
S-TEACH-03	Teacher Material Feed	Teacher	FR-T-04	P1
S-MOES-01	MoES Analytics Dashboard	MoES Official	FR-M-03	P1
S-MOES-02	Curriculum Content Manager	MoES Official	FR-M-02	P1
S-MOES-03	Report Generator	MoES Official	FR-M-04	P2
S-SHARED-01	Notification Centre	All roles	FR-L-08	P2
S-SHARED-02	Profile & Settings	All roles	FR-L-10, FR-L-11	P2

0.3 Frontend Architecture Decisions

The UI is a standalone React SPA communicating with the Django/DRF backend exclusively through REST API calls and WebSocket. The following frontend architecture applies across all screens:

Concern	Decision	Rationale
Framework	React 18 + Vite	SPA, fast HMR, tree-shaking, PWA plugin support
Routing	React Router v6 (createBrowserRouter)	Nested layouts, role-based route guards, lazy loading
State management	Zustand (global) + React Query (server state)	Zustand for auth/UI state; React Query for all API data, caching, invalidation
Authentication	JWT — access token in memory, refresh token in httpOnly cookie	Secure: XSS cannot steal refresh token; access token short-lived (5 min)
Auth flow	Axios interceptor refreshes access token silently on 401	Seamless session continuation without forcing logout
API layer	Axios instance with baseURL, auth interceptor, error boundary	Centralised request/response handling, toast on error
Offline support	Vite PWA plugin (Workbox) + IndexedDB via idb library	Service Worker caches lesson bundles; idb for structured offline reads
WebSocket	native WebSocket + custom useWebSocket hook	Django Channels real-time collab and notifications
Forms	React Hook Form + Zod validation	Performant, schema-based validation, no re-render on every keystroke
Component library	Custom design system (no Bootstrap/MUI)	Platform-specific identity; avoids generic visual language
CSS	CSS Modules + CSS custom properties	Scoped styles, no class collision, theming via variables
Accessibility	WCAG 2.1 AA — focus rings,	NFR-AC-02: functional on low-end Android; shared

Concern	Decision	Rationale
	ARIA labels, colour contrast ≥4.5:1	devices
Code splitting	React.lazy + Suspense per route	Reduces initial bundle; critical for 3G (NFR-P-01)
Image optimisation	WebP format, responsive srcset, lazy loading	NFR-P-03: low-bandwidth delivery

0.4 Layout System

All screens share a consistent shell layout composed of four regions:

- Top Navigation Bar (64px desktop / 56px mobile) — global nav, role-aware links, AI Tutor link, notifications, avatar
- Left Sidebar (240px desktop, collapses to icon rail at 64px on tablet, hidden on mobile) — section navigation
- Main Content Area (fluid, max-width 1200px, centered) — primary content
- Contextual Right Panel (320px, optional, slides in) — AI Tutor chat, filters, post details

Mobile layout collapses left sidebar into a bottom navigation bar (5 tabs max). Right panel becomes a full-height bottom sheet.

0.5 Contextual AI Tutor Interaction Pattern

This is the most architecturally distinctive UI decision in the platform. The AI Tutor operates in three modes simultaneously:

1. Dedicated Page (S-AI-01) — full-screen AI chat interface accessible via "AI Tutor" link in the top navigation bar. Behaves like a persistent intelligent tutor with full session history.
2. Content Card AI Button — every lesson card, post card, and teacher material card carries a teal AI icon button. On hover/tap, a smart dropdown appears showing 3–4 contextually generated suggested questions about that specific content (e.g. "Explain this concept", "Give me a quiz on this", "Summarise for S3 level"). The learner selects a suggestion or types a custom question. On submit, the content payload + question are sent to the AI Tutor API.
3. Inline AI Panel — when invoked from a content card, the right contextual panel slides in (desktop) or a bottom sheet rises (mobile) showing the AI response inline, without navigating away from the current page. The learner can continue the conversation in that panel or open the full AI page.

This pattern is analogous to Grok in X but more structured: the AI is always contextually aware of what content the learner is currently viewing, without requiring the learner to manually copy-paste content into a chat box.

Part 1 — Design System Foundation Prompt

Paste this prompt first in every new Google Stitch session. It establishes the design system that all subsequent screen prompts build on.

PROMPT G-DS-01 — Design System Foundation (Paste First in Every Session)

Design a complete UI design system for the CBC Digital Learning Platform — Uganda's national secondary school edtech platform serving learners (S1–S6), teachers, and Ministry of Education officials. IDENTITY & TONE A platform that is simultaneously institutional and youth-friendly. Authoritative but approachable. The visual language should feel like a modern African educational institution — not a Western SaaS tool, not a generic LMS. COLOUR SYSTEM - Primary: Deep Institutional Blue #1B4F72 - Primary Light: #2980B9 (interactive states) - Secondary: Uganda Green #1E8449 - Secondary Light: #27AE60 - Accent / CTA: Warm Amber #D4A017 - AI Tutor Brand: Deep Teal #0E6655 - Surface 1 (background): #F8F9FA - Surface 2 (cards): #FFFFFF with border #E8EAED - Surface 3 (sidebar/nav): #1B2631 - Text Primary: #1A252F - Text Secondary: #5D6D7E - Text Inverse: #FFFFFF - Success: #27AE60 | Error: #C0392B | Warning: #F39C12 | Info: #2980B9 TYPOGRAPHY - Display font: DM Serif Display (Google Fonts) — all headings H1–H3 - Body font: Plus Jakarta Sans (Google Fonts) — all body text, labels, UI - Mono font: JetBrains Mono — AI responses, code, lesson body text - Scale: 12 / 14 / 16 / 18 / 24 / 32 / 40px. Line height 1.5 body, 1.2 headings - Font weights: 400 body, 500 medium labels, 600 semibold, 700 bold headings SPACING & GRID - Base unit: 8px. All spacing multiples of 8 (8, 16, 24, 32, 40, 48, 64, 96) - Page max-width: 1200px, centered. Side padding: 24px mobile, 48px desktop - 12-column grid. Gutters: 24px desktop, 16px mobile - Card border-radius: 12px. Button border-radius: 8px. Input border-radius: 8px. Pill: 999px ELEVATION (shadows) - Level 1 (cards): 0 1px 3px rgba(0,0,0,0.08), 0 1px 2px rgba(0,0,0,0.04) - Level 2 (dropdowns, tooltips): 0 4px 16px rgba(0,0,0,0.12) - Level 3 (modals, sheets): 0 8px 32px rgba(0,0,0,0.18) COMPONENT SPECIFICATIONS Buttons: Primary (filled #1B4F72, white text), Secondary (outlined #1B4F72), Accent (filled #D4A017, dark text), Ghost (transparent), Danger (filled #C0392B). All 40px height desktop / 44px mobile minimum. Input fields: 48px height, #E8EAED border, 2px focus ring #2980B9, floating label pattern. Cards: white background, 12px radius, Level 1 shadow, 24px padding. AI Tutor button (on cards): 32px circular teal #0E6655 button, brain/sparkle icon, appears bottom-right of every content card. NAVIGATION SHELL Top bar: 64px, #1B2631 background, white text. Left: platform logo + wordmark. Centre: search bar (desktop). Right: AI Tutor link (teal badge), notifications bell, user avatar with role chip. Left sidebar: 240px, #1B2631. Section links with icons. Collapses to 64px icon rail on tablet. Mobile: top bar persists. Bottom navigation bar (5 tabs) replaces sidebar. OFFLINE INDICATOR Persistent banner (32px, amber #F39C12 background) below top nav when device is offline. Text: "You are offline. Only downloaded lessons are available." AI TUTOR VISUAL IDENTITY All AI-related UI elements use teal #0E6655 as brand colour. AI responses use a distinct left-border-accented bubble in teal. AI icon: a stylised brain/sparkle glyph. Label: "AI Tutor" always shown. ACCESSIBILITY WCAG 2.1 AA. All interactive elements minimum 44×44px touch target. Colour contrast ratio ≥ 4.5:1 for body text. Focus indicators: 2px solid #2980B9 outline. Skip navigation link. Screen reader ARIA labels on all icon buttons.

Part 2 — Authentication Screens

SRS Source: FR-L-01, FR-L-02, FR-T-01, FR-M-01 | Use Cases: UC-U-01, UC-U-02, UC-U-03 | NFR-SEC-03

Screen S-AUTH-01 — Registration

PROMPT G-AUTH-01 — Registration Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Registration page. LAYOUT Two-column split layout (desktop). Left column (45%): full-height brand panel with #1B4F72 background, platform logo centred top, large DM Serif Display headline "Start Learning Uganda's Curriculum", a short subtext about the platform, and three illustrated feature callout cards at the bottom (CBC Content, AI Tutor, Learn Offline). Right column (55%): white background, registration form. Mobile: single column. Brand panel collapses to a slim 120px header strip with logo + tagline only. FORM — RIGHT COLUMN Title: "Create Your Account" (DM Serif Display, 32px). Subtitle: "Join thousands of learners across Uganda." Step indicator: 2-step pill progress bar at the top (Step 1: Account Details | Step 2: Role Setup). STEP 1 FIELDS (all with floating labels, 48px height): - Full Name (text input) - Username (text input, inline availability check with green tick / red X) - Email Address (optional, text input) - Password (password input, strength meter bar below — Weak/Fair/Strong in red/amber/green) - Confirm Password Role selector: Three large pill cards side by side — Learner / Teacher (each self-register). Cards have icon, role name, short description. Selected state: #1B4F72 border + light blue fill. Primary CTA button: "Continue to Step 2" (full width, #D4A017 amber, dark text, 48px height). STEP 2 FIELDS (conditional based on role): If Learner: School search (searchable dropdown), Class Level selector (S1–S6 as large pill buttons in a 3-column grid). If Teacher: School search, subjects taught (multi-select tag input with CBC subjects as options). Final CTA: "Create Account" (full width, #1B4F72 primary, white text). Below CTA: "Already have an account? Log in" link. MICRO-INTERACTIONS Username availability: debounced API check, inline spinner then tick/cross. Password strength: animated bar fills left to right. Role card selection: subtle scale(1.02) + border colour transition on select. Form step transition: slide-left animation between Step 1 and Step 2.

Screen S-AUTH-02 — Login

PROMPT G-AUTH-02 — Login Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Login page. LAYOUT Centred card layout (not split). Vertically and horizontally centred on the page. Page background: subtle diagonal geometric pattern using #1B4F72 at 4% opacity on #F8F9FA. Card: white, 480px wide (desktop), 100% width – 32px on mobile. 12px border-radius. Level 3 shadow. 48px padding. CONTENT Top: platform logo (80px) centred above card. Card header: "Welcome Back" (DM Serif Display, 32px, centre-aligned). Subtext: "Log in to continue learning." Role indicator tabs: three tabs at top of form — Learner | Teacher | MoES Official. Selecting a tab does not change the form fields but sets the role context visually and sends role hint to the API. Active tab: #1B4F72 bottom border, bold label. FIELDS - Username (text input, 48px, floating label) - Password (password input, toggle show/hide eye icon on right) Primary CTA: "Log In" (full width, #1B4F72, 48px, white text, rounded 8px). Below button: "Forgot Password?" right-aligned link (not implemented in prototype, shown greyed with tooltip "Coming soon"). Divider line with "or" text. Below divider: "Don't have an account? Register" link. ERROR STATES Inline error below each field in red #C0392B with warning icon. If credentials fail: red dismissible alert banner above the form "Incorrect username or password. Please try again." LOADING STATE On submit: button shows spinner, text changes to "Logging in...", button disabled. OFFLINE NOTICE If device is offline when login page loads: amber banner above card "You are offline. Login requires internet connection."

Part 3 — Learner Screens

SRS Source: FR-L-03 to FR-L-11 | Use Cases: UC-L-04 to UC-L-13 | NFR-P-01, NFR-O-01 to NFR-O-03

Screen S-LEARN-01 — Learner Home / Feed

PROMPT G-LEARN-01 — Learner Home Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Learner Home page — the first screen a learner sees after login. SHELL Apply the full navigation shell: dark top bar (#1B2631) with logo left, search bar centre, AI Tutor nav link (teal, brain icon, subtle pulse animation), notification bell (badge count), and learner avatar + class level chip (e.g. "S3") on the right. Left sidebar open by default on desktop showing: Home, Subjects, My Library (offline), Progress, Knowledge Wall, Collaborative Sessions. Active link has left #D4A017 amber border + #1B4F72 text. PAGE LAYOUT Two-column: main content (8 cols) + right panel (4 cols) on desktop. Single column stacked on mobile. MAIN CONTENT — TOP SECTION Personalised welcome banner: "Good morning, [Name]" (DM Serif Display, 24px) with learner's class level and school badge. Amber left-border accent card. Inside: a motivational CBC progress stat — "You have attained 4 of 12 competencies this term" with a thin progress bar. CONTINUE LEARNING SECTION Section heading "Continue Learning" with "View All" link right-aligned. Horizontal scroll row of Lesson Cards (3 visible on desktop, 1.5 on mobile — signals scrollability). LESSON CARD ANATOMY (to be reused across the platform): - 240×160px thumbnail area (subject colour gradient if no image, with subject icon centred) - Subject chip (coloured pill, e.g. "Biology" in green) - Class level chip ("S3") - Lesson title (Plus Jakarta Sans, 16px, semibold, 2-line clamp) - Competency tag below title (grey, 12px) - Bottom row: download icon button (left), AI Tutor button (teal circular 32px, bottom-right, brain/sparkle icon) - Offline badge (amber "Downloaded" pill) if lesson is in offline cache AI TUTOR BUTTON BEHAVIOUR ON CARD (critical pattern): On hover (desktop) or long-press (mobile) on the teal AI button: a dropdown card appears anchored above the button. Dropdown contains: - Header: "Ask AI Tutor about this lesson" (teal, 13px bold) - 3 suggested question pills (auto-generated from lesson title + competency): e.g. "Explain this concept in simple terms" | "Give me 3 exam questions on this" | "Summarise for S3 level" - A text input "Or ask your own question..." with send button On selection or submit: right panel slides in (desktop) or bottom sheet rises (mobile) with AI Tutor response. User does not navigate away. SUBJECTS SECTION Section heading "Browse by Subject" with horizontal scroll row of Subject Cards. Subject Card: square 120×120px, subject colour background, white icon centred, subject name below in white. CBC subjects: Mathematics, Biology, Chemistry, Physics, History, Geography, English, Kiswahili, Agriculture, etc. RIGHT PANEL — TODAY'S HIGHLIGHTS Card: "Today on the Knowledge Wall" showing 2 recent peer posts (compact). "View All" link. Card: "Your Progress" — circular progress ring showing overall competency attainment percentage. Card: "Active Sessions" — if any collaborative sessions are live, shown here with "Join" CTA.

Screen S-LEARN-03 — Lesson Reader

PROMPT G-LEARN-03 — Lesson Reader Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Lesson Reader — the primary content consumption screen. LAYOUT Reading-optimised layout. Left sidebar collapses to icon rail automatically on this screen to maximise reading width. Max content width: 720px, centred. Right panel available for AI Tutor (slides in on demand without disturbing reading). TOP LESSON HEADER (full-width, #1B2631 background, 120px height) Breadcrumb: Home > Subjects > Biology > S3 > [Lesson Title] (white, 13px) Lesson title (DM Serif Display, 32px, white) Metadata row: Subject chip | Class Level chip | Competency tag | "NCDC" source badge | Reading time estimate Action row (right-aligned): Download for

Offline button (cloud-down icon + text), Share button, AI Tutor button (teal, prominent, "Ask AI Tutor" label + brain icon) LESSON CONTENT AREA Body rendered in JetBrains Mono-accented sections with Plus Jakarta Sans body text. Section headings: DM Serif Display, 24px, #1B4F72. Body text: Plus Jakarta Sans, 16px, #1A252F, line-height 1.7 (optimised for reading). Images: full-width within content column, 8px border-radius, lazy-loaded. Key term callout boxes: left border accent in #27AE60, light green background tint. Definition boxes: left border in #2980B9, light blue background. LESSON COMPLETION SECTION (bottom of content) "Mark as Complete" large button — #1E8449 green, full width of content column. Below: "Competency Progress Update" — shows which competency this lesson contributes to and current attainment level pill (NOT_STARTED → IN_PROGRESS → ATTAINED). NAVIGATION Previous / Next lesson buttons at bottom-left and bottom-right. Floating "Back to Top" button appears after scroll > 300px. AI TUTOR SIDE PANEL (slides in from right, 360px wide, does not cover content on desktop) Panel header: teal #0E6655 background, "AI Tutor" label + brain icon, X close button. Inside: a pre-filled message "I'm reading: [Lesson Title]. " with cursor ready. 3 suggested question pills about this specific lesson pre-loaded. Chat thread below: user messages right-aligned (blue bubble), AI responses left-aligned (teal left-border, JetBrains Mono font for any structured content, Plus Jakarta Sans for prose). "Out of scope" state: amber card "This question is outside the CBC knowledge base. Try rephrasing within the curriculum topic." Input bar fixed at panel bottom: textarea + send button. OFFLINE STATE If reading offline: amber strip below top header "You are reading an offline copy downloaded on [date]." AI Tutor button disabled with tooltip "AI Tutor requires internet connection."

Screen S-LEARN-04 — Offline Library

PROMPT G-LEARN-04 — Offline Library Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Offline Library page — all lessons the learner has downloaded for offline access. LAYOUT Standard shell. Page header with offline cloud icon + "My Offline Library" title (DM Serif Display, 32px). OFFLINE STATUS BANNER If currently offline: full-width amber banner at top of content "You are offline. Showing your downloaded lessons." with WiFi-off icon. If online: green "Connected" small pill near header, no banner. STORAGE INDICATOR Card at top: "Device Storage Used" — horizontal segmented bar showing KB used vs available. Colour: #1B4F72 fill on #E8EAED background. Text: "124 MB of downloaded lessons" with "Manage Storage" link. LESSON GRID All downloaded lessons in a responsive grid (3 cols desktop, 2 cols tablet, 1 col mobile). Each card: same Lesson Card anatomy as S-LEARN-01 but with: - "Downloaded" amber badge top-right - "Downloaded [date]" metadata below title - Remove from library trash icon button (bottom-left, ghost style, #C0392B on hover) - Storage size shown (e.g. "1.2 MB") EMPTY STATE If no lessons downloaded: centred illustration of a cloud with down arrow (use CSS/SVG, no external image), heading "No lessons downloaded yet", subtext "Download lessons while connected to read them anywhere.", CTA button "Browse Lessons" (#1B4F72).

Screen S-LEARN-05 — Competency Progress Dashboard

PROMPT G-LEARN-05 — Learner Progress Dashboard

Using the CBC Digital Learning Platform design system (G-DS-01), design the Learner Progress Dashboard — showing CBC competency attainment per subject. LAYOUT Standard shell. Page header: "My Progress" (DM Serif Display, 32px). Class level + term badge right-aligned. OVERVIEW ROW — 3 STAT CARDS (equal width, horizontal row) Card 1: Large circular progress ring (SVG, animated on load). Centre: percentage (e.g. "42%"). Below ring: "Competencies Attained". Colour: #1E8449. Card 2: "X Attained" with a green checkmark icon, large number. Card 3: "X In Progress" with an amber spinner icon, large number. SUBJECT ACCORDION LIST Each CBC subject is a collapsible accordion row. Collapsed state: Subject name (semibold

16px) | Competency count (e.g. 3/8 attained) | Mini progress bar | expand chevron. Subject row background tinted with subject colour at 8% opacity. Expanded state reveals a list of competency rows, each showing: - Competency name (14px, regular) - Attainment pill: NOT STARTED (grey) | IN PROGRESS (amber) | ATTAINED (green) — colour-coded, pill shape - Last updated date (12px, #5D6D7E) - "Learn More" link (opens related lessons) PROGRESS TIMELINE (bottom section) Horizontal timeline chart (CSS-based bar chart) showing competencies attained per month. #1B4F72 bars. Month labels below. Title: "Attainment Over Time".

Part 4 — Knowledge Wall (Posts & Collaboration)

SRS Source: FR-L-06, FR-L-07, FR-L-08 | Use Cases: UC-L-08, UC-L-09, UC-L-10, UC-L-11 | NFR-SEC-05

Screen S-POST-01 — Knowledge Wall Feed

PROMPT G-POST-01 — Knowledge Wall Feed

Using the CBC Digital Learning Platform design system (G-DS-01), design the Knowledge Wall — the national peer knowledge sharing feed. LAYOUT Two-column: main feed (8 cols) + right panel (4 cols, desktop). Single column mobile. PAGE HEADER "Knowledge Wall" (DM Serif Display, 32px). Subtext: "Insights, research, and ideas from learners across Uganda." Filter row below header: Subject filter dropdown | Class Level filter | Sort (Latest / Most Reactions / My Posts) | Create Post button (amber CTA, right-aligned, "+ Post" label). POST CARD ANATOMY (vertical stack in main feed) White card, 12px radius, Level 1 shadow, 24px padding, full width. Top row: learner avatar (40px circle, initials fallback) | learner name (semibold 15px) | school name + region (13px, grey) | class level badge | time ago (13px, right-aligned). Visibility indicator: small pill top-right of card — PRIVATE (grey lock), PEERS (blue people icon), TEACHERS (purple graduation icon), PUBLIC (green globe). Content body: post text (16px, 4-line clamp with "Read more" expand). CBC tags row: subject pill + competency pill (teal outline, small). Action row (bottom of card): - Reaction buttons: thumbs-up LIKE (with count), share icon SHARE (with count). On click: icon animates (scale bounce), count increments optimistically. - Comment button: speech bubble icon + count. On click: inline comment thread expands below card (accordion, does not navigate). - AI Tutor button: teal circular 32px brain icon, bottom-right. Hover/tap shows suggested question dropdown: "Analyse this insight", "Find related CBC content", "Ask a follow-up question on this post", custom input. INLINE COMMENT THREAD (expands below post card on comment button click) Indented 16px from card left edge. Each comment: avatar + name + text + time. Input bar at bottom: avatar + textarea + "Post Comment" button. RIGHT PANEL "Trending This Week" — top 5 most-reacted posts as compact list items. "Active Learners" — count of learners online nationally (from API, or "—" if unavailable). "Filter by Competency" — tag cloud of active competencies.

Screen S-POST-02 — Create Post

PROMPT G-POST-02 — Create Post Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Create Post page — where learners write and publish knowledge contributions. LAYOUT Centred single-column, max-width 720px. Clean, distraction-free writing environment. PAGE HEADER "Share Your Insight" (DM Serif Display, 32px). Subtext: "Contribute to Uganda's collective knowledge base." FORM Title field (optional): text input, large 20px placeholder "Give your post a title (optional)" Content area: large textarea, minimum 200px height, grows with content. Placeholder: "Share what you've learned, a research finding, or an idea connected to your CBC studies..." Character count: bottom-right of textarea, turns amber at 80% of limit, red at limit. CBC TAGGING SECTION "Tag your post" subheader (14px semibold). Subject dropdown (searchable) + Class Level dropdown. Competency multi-select: appears after subject is chosen, shows competency pills to select, max 3. VISIBILITY SELECTOR "Who can see this?" label (14px semibold). Four pill-button options in a row: 🔒 Private | 👥 Peers | 🧑🏫 Teachers | 🌍 Public. Each shows a description tooltip on hover. Default selected: Private (grey filled). Selected state: #1B4F72 filled + white text. ACTION ROW Cancel link (left) | "Preview" button (secondary outlined) | "Post" CTA (amber filled, right). PREVIEW MODE On "Preview" click: form fields collapse and post renders exactly as it would appear on the Knowledge Wall, using the Post Card component. "Edit" and "Publish" buttons replace the preview toggle.

Part 5 — AI Tutor Dedicated Page

SRS Source: FR-L-05, FR-AI-01 to FR-AI-06 | Use Cases: UC-L-07, UC-AI-01 to UC-AI-05 | NFR-P-02, NFR-SEC-05

PROMPT G-AI-01 — AI Tutor Dedicated Page (Full Screen)

Using the CBC Digital Learning Platform design system (G-DS-01), design the AI Tutor dedicated page — a full-featured AI tutor interface accessible from the top navigation "AI Tutor" link. This screen functions like an intelligent academic assistant deeply embedded in the platform — analogous to how Grok is integrated in X (formerly Twitter), but purpose-built for CBC curriculum learning. The AI is not a generic chatbot. It is curriculum-aware, always operating within CBC subject boundaries. LAYOUT Two-panel layout. Left panel (320px, fixed): session history sidebar. Right panel (fluid): active chat interface. Mobile: left panel collapses, accessible via hamburger icon inside the AI page header. LEFT PANEL — SESSION HISTORY SIDEBAR Background: #1B2631 (same as main nav, unified dark family). Header: teal brain icon + "AI Tutor" wordmark in teal #0E6655. Small badge: "CBC Knowledge Base v1.0". "New Session" button: teal filled, full width, "+ New Conversation" label. Session list below: each past session shown as a compact row: - Session title (auto-generated from first query, truncated 30 chars) - Subject chip (coloured small pill) - Date (relative, "2 hours ago") - Active session has left teal border accent. Footer: "Sessions are private to you" (12px, grey, lock icon). Links to Privacy Settings. RIGHT PANEL — CHAT INTERFACE TOP BAR (inside AI panel, not the global nav) Teal #0E6655 background, 56px height. Left: current session subject chip + "AI Tutor" label. Right: "Clear Session" ghost button | "Expand to Full Page" icon (if invoked from inline panel, this navigates to this page). WELCOME STATE (new session, no messages yet) Centred vertically in the chat area. Large teal brain icon (64px SVG). Heading: "What do you want to learn today?" (DM Serif Display, 28px). Subtext: "Ask me anything about your CBC subjects. I'm grounded in Uganda's national curriculum." CBC SUBJECT QUICK-START GRID 3×3 grid of subject pills (same subjects as platform): each a card with subject icon + name. Clicking pre-fills the context and shows subject-specific starter questions. SUGGESTED STARTER QUESTIONS (shown below subject grid) 4 pill buttons with example questions: "Explain photosynthesis for S3 Biology", "What are the competencies for S4 Mathematics?", "Quiz me on Uganda's history topics", "Summarise the water cycle." ACTIVE CHAT STATE Chat thread area (scrollable, padding 24px): User message: right-aligned, #1B4F72 filled bubble, white text, Plus Jakarta Sans 15px. Avatar right of bubble. AI response: left-aligned. Teal #0E6655 left border (4px) on a white card with very light teal background tint. AI brain icon top-left of card. "AI Tutor" label in teal bold above response text. Response text in Plus Jakarta Sans 15px, line-height 1.6. Any structured output (lists, steps, definitions) uses JetBrains Mono 14px in a lightly backgrounded inner block. Typing indicator: three animated teal dots while AI is processing. Timestamp: small grey text below each message pair. OUT-OF-SCOPE STATE AI response card: amber left border (4px). Amber brain icon. Message: "This question appears to go beyond Uganda's CBC curriculum scope. I can best help with questions related to your CBC subjects. Would you like to rephrase within [Subject]?" INPUT AREA (fixed at panel bottom, never scrolls away) White bar, Level 2 shadow above. Left: subject context chip (shows current subject context, clickable to change, e.g. "Biology S3"). Textarea: "Ask your question..." placeholder, auto-expands to 4 lines max. Right: Send button (teal circle, arrow icon, 44px). Disabled when textarea empty. Below input: small "AI responses are based on Uganda's NCDC curriculum. Always verify with your teacher." disclaimer in 12px grey. CONTENT-CONTEXT STATE (when invoked from a content card) Pre-filled banner above input area: light teal background card showing "Context: [Lesson/Post title]" with an X to remove context. Input pre-filled with selected suggestion or custom question. This state makes clear to the learner that the AI is reading the content they selected.

Part 6 — Teacher Screens

SRS Source: FR-T-01 to FR-T-04 | Use Cases: UC-T-04 to UC-T-09 | Stage C: Teacher, TeacherMaterial entities

Screen S-TEACH-01 — Teacher Home

PROMPT G-TEACH-01 — Teacher Home Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Teacher Home page. SHELL DIFFERENCES FROM LEARNER Same top navigation shell. Left sidebar links differ: Dashboard, My Materials, Upload Material, My Learners, AI Tutor. Role chip in avatar dropdown shows "Teacher". PAGE CONTENT Welcome card (same amber-accented pattern as learner home but contextualised): "Welcome back, [Teacher Name]" | "[School Name]" badge | "Subjects: Mathematics, Physics" chips. STAT ROW — 4 METRIC CARDS Materials Uploaded (count, blue), Shared with Learners (count, green), Learner Views This Week (count, amber), Subjects Active (count, grey). Each card: large number (DM Serif Display, 40px), label below (14px), icon top-right. RECENT MATERIALS SECTION Section heading "Your Recent Materials" | "View All" link | "Upload New" amber CTA button (right). Table layout (desktop) / card stack (mobile). Table columns: Material Title | Subject | Class Level | Resource Type | Shared | Uploaded | Actions. Actions: Edit pencil icon | Toggle share switch | Delete trash icon (with confirmation modal). UPLOAD NEW MATERIAL CTA Large dashed-border card at bottom of materials list: "+ Upload New Material" centred, cloud-up icon, subtext "Share study guides, past papers, notes with your learners."

Screen S-TEACH-02 — Upload Material

PROMPT G-TEACH-02 — Upload Material Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Upload Material page. LAYOUT Centred single-column, max-width 720px. Clean form with clear section grouping. PAGE HEADER "Upload Learning Material" (DM Serif Display, 32px). Breadcrumb: Dashboard > My Materials > Upload New. FORM SECTIONS (use card grouping — each section in a white card with section heading) SECTION 1: MATERIAL DETAILS Title input (required, floating label) Description textarea (optional, 4 rows) Resource Type: pill selector — Study Guide | Past Paper | Notes | Worksheet | Other. Selected: #1B4F72 filled. SECTION 2: CBC TAGGING (required) Subject dropdown (searchable, required) Class Level multi-select pills (S1–S6, can select multiple, required) Competency association (appears after subject chosen, optional multi-select tag input) SECTION 3: FILE UPLOAD Large dashed drop zone (full width, 180px height, dashed #1B4F72 border at 40% opacity). Icon: cloud-up SVG. Text: "Drag and drop your file here" | "or Browse Files" link (underlined). Accepted formats: PDF, DOCX, PPTX, MP4 (max 50MB). Shown as small grey chips below drop zone. Upload progress: replaces drop zone on file selection — filename + progress bar + cancel X button. Multiple files: after first upload, "+ Add Another File" link appears. SECTION 4: SHARING SETTINGS "Share with learners immediately" toggle switch (default OFF, label explains: "Turn on to make this material visible to learners in your class levels now. You can change this later."). ACTION ROW Cancel link | "Save as Draft" secondary button | "Upload Material" primary CTA (amber filled). SUCCESS STATE Full page replaced by success card: green checkmark animation, "Material Uploaded Successfully", material title shown, two CTA buttons: "Upload Another" | "View My Materials".

Part 7 — MoES / NCDC Official Screens

SRS Source: FR-M-01 to FR-M-04 | Use Cases: UC-M-03 to UC-M-08 | NFR-S-01, NFR-S-03

Screen S-MOES-01 — National Analytics Dashboard

PROMPT G-MOES-01 — MoES/NCDC Analytics Dashboard

Using the CBC Digital Learning Platform design system (G-DS-01), design the National Analytics Dashboard for MoES/NCDC officials. SHELL DIFFERENCES Left sidebar: Dashboard, Curriculum Manager, Reports, Knowledge Base, User Management. Role chip: "MoES Official" in a distinct dark gold (#9A7D0A) badge. DASHBOARD PHILOSOPHY This is a data-rich governance dashboard. Layout is denser than learner screens. Information hierarchy: national → regional → school → subject → competency. PAGE HEADER "National Learning Dashboard" (DM Serif Display, 32px). Real-time badge: green dot + "Live Data" label (right-aligned). Last updated timestamp. TOP METRIC STRIP (6 equal cards in a horizontal row) Total Registered Learners | Active This Week | Lessons Accessed Today | AI Tutor Queries Today | Competencies Attained (Total) | Teachers Active. Each: large number (DM Serif Display, 40px), delta arrow (up/down, green/red) showing % change from last week. ROW 2 — TWO CHARTS (side by side) Left chart (7 cols): "Platform Engagement — Last 30 Days" — Area line chart. X-axis: dates. Y-axis: active learners. Two lines: Lesson Access (blue) and AI Tutor Usage (teal). Fill area below each line with matching colour at 15% opacity. Legend below chart. Right chart (5 cols): "Competency Attainment by Subject" — Horizontal bar chart. Each CBC subject is a bar. Bars coloured by attainment rate: green >70%, amber 40–70%, red <40%. Percentage labels at bar end. ROW 3 — REGIONAL MAP + SUBJECT TABLE (side by side) Left (7 cols): "Engagement by Region" — Uganda regions as a simple block/tile map (CSS grid or SVG), each region coloured by engagement intensity (blue scale, darker = more active). Hover tooltip: region name + active learner count + top subject. Right (5 cols): Top 5 Competencies table — columns: Competency | Subject | Attainment % | Trend arrow. Sortable headers. FILTER BAR (below page header, above metric strip) Date range picker | Region dropdown | Class Level multi-select | Subject dropdown | "Apply Filters" button. Filters affect all charts and metrics on the page simultaneously. REPORT CTA Persistent bottom-right floating button: "Generate Report" (#1B4F72 filled, document icon, 48px height).

Part 8 — Shared Component Prompts

These components appear across multiple screens and roles. Generate each independently in Stitch, then compose them into screens.

Component: Notification Centre

PROMPT G-COMP-01 — Notification Centre (Dropdown + Full Page)

Using the CBC Digital Learning Platform design system (G-DS-01), design the Notification Centre. TWO STATES TO DESIGN: STATE 1: DROPDOWN (triggered from bell icon in top nav) Anchored below the notification bell icon, 380px wide, max-height 520px, scrollable. Header row: "Notifications" (semibold 16px, left) | "Mark all read" link (right, teal text). Notification types with distinct left-border colours: - Post reaction: blue border — "[Name] liked your post about [topic]" - Comment: green border — "[Name] commented on your post" - New lesson: blue border — "New S3 Biology lesson added: [title]" - Collaborative session: amber border — "[Name] started a session: [title]" - Competency attained: green border — "You attained: [Competency Name]!" Each notification: avatar + text (2-line clamp) + time-ago + unread blue dot (if unread). Footer: "View All Notifications" full-width link. STATE 2: FULL NOTIFICATION PAGE Same items in a full-page list grouped by date: Today / Yesterday / This Week / Earlier. Filter tabs: All | Unread | Lessons | Posts | Sessions.

Component: Profile & Settings

PROMPT G-COMP-02 — Profile & Settings Page

Using the CBC Digital Learning Platform design system (G-DS-01), design the Profile & Settings page. LAYOUT Two-column: left settings nav (240px) + right content area (fluid). LEFT NAV LINKS (with icons) My Profile | Account & Security | Post Visibility Defaults | Offline Storage | Notifications | About PROFILE SECTION (default view) Avatar circle (96px, initials fallback, edit overlay button on hover shows camera icon). Name (DM Serif Display, 24px) | Username chip | Role badge | School name | Class level badge. Edit Profile button (outlined secondary). Editable fields below: Full Name | Email (optional) | School (read-only) | Class Level. Save Changes button (primary). POST VISIBILITY DEFAULTS SECTION "Default visibility for new posts" — four pill options (Private/Peers/Teachers/Public). Subtext explains each. "Default visibility for AI sessions" — toggle: Private only (always locked, with explanation: "Your AI sessions are always private and cannot be shared."). ACCOUNT & SECURITY SECTION Change Password: current password + new password + confirm fields. Show/hide toggles. Session Management: list of active sessions with device name + last active time + "Log out" link per session.

Part 9 — Responsive & Offline State Prompts

PROMPT G-RESP-01 — Mobile Navigation Shell

Using the CBC Digital Learning Platform design system (G-DS-01), design the mobile navigation shell for screens narrower than 768px (Android smartphone — 360px to 430px width primary range). MOBILE SHELL COMPONENTS: TOP BAR (56px height, #1B2631) Left: hamburger menu icon (opens a full-height drawer overlay for secondary navigation). Centre: platform logo (icon only, no wordmark — space constrained). Right: AI Tutor brain icon (teal, 44px tap target, navigates to S-AI-01), notification bell with badge, avatar (36px circle). BOTTOM NAVIGATION BAR (56px height, white, Level 3 shadow above, fixed to bottom) 5 tabs: Home (house icon) | Subjects (book icon) | AI Tutor (brain icon, teal accent, slightly larger to signify primary action) | Knowledge Wall (people icon) | Progress (chart icon). Active tab: #1B4F72 icon fill + label visible (13px, semibold). Inactive: grey icon, no label. Unread badges on applicable tabs. LEFT DRAWER (slides in from left, 280px, 100vh, #1B2631) Same links as desktop sidebar but full-height overlay. Learner avatar + name at top. Tap outside or swipe-left to close. BOTTOM SHEET (right panel equivalent on mobile, slides up from bottom) Used for AI Tutor inline panel, post details, filters. Max-height 85vh. Drag handle (4px pill) at top centre. Dismisses by dragging down or tapping background scrim.

PROMPT G-RESP-02 — Offline Mode States

Using the CBC Digital Learning Platform design system (G-DS-01), design all offline-specific UI states. STATE 1: OFFLINE GLOBAL BANNER 32px amber (#F39C12) strip pinned below the top navigation bar (not inside it). Left: WiFi-off icon (16px, white). Text: "You are offline — only downloaded lessons are available." Right: "My Library" link in white underlined. This banner must appear on ALL pages when the device is offline. It must never obscure interactive content. STATE 2: LESSON CARD OFFLINE STATES Downloaded + Online: green "Downloaded" badge top-right. Normal AI button. Downloaded + Offline: amber "Offline" badge top-right. AI button grey + disabled tooltip "Requires internet". Not downloaded + Offline: card shows 40% opacity, "Not Available Offline" overlay label centred on thumbnail. Download button replaced by lock icon. STATE 3: AI TUTOR OFFLINE STATE On S-AI-01 (AI Tutor page) when offline: full chat area replaced by a centred state card. Brain icon in grey (not teal). Heading: "AI Tutor Unavailable" (grey, DM Serif Display, 24px). Subtext: "AI Tutor requires an internet connection. You can still access your downloaded lessons." CTA button: "Go to My Library" (#1B4F72). Session history sidebar still visible and navigable (cached locally). STATE 4: FEATURE DISABLED TOOLTIP (applies to Posts, Collab Sessions, etc.) Any feature requiring internet when offline: button/link shows grey disabled state. On hover/tap: tooltip appears above element: "This feature requires internet connection." (12px, dark grey, arrow pointing to button).

Part 10 — Frontend Architecture Specification

This section defines the complete technical frontend architecture for the React SPA. This is not a Stitch prompt — it is the developer specification derived from the UI design to guide implementation.

10.1 Project Structure

```
src/
  api/              # Axios instance, interceptors, endpoint functions
  assets/           # Static images, fonts, icons
  components/       # Shared reusable components (Button, Card, Input, etc.)
  features/         # Feature-scoped modules
    auth/           # Login, Register, useAuth hook
    learner/        # Home, LessonReader, Progress, OfflineLibrary
    posts/          # Feed, CreatePost, PostCard, Comments
    ai-tutor/       # AITutorPage, AIPanel, AIButton, AIDropdown
    teacher/        # TeacherHome, UploadMaterial, MaterialFeed
    moes/           # AnalyticsDashboard, CurriculumManager, Reports
    shared/         # Notifications, Profile, Settings
  hooks/            # Custom hooks (useWebSocket, useOffline, useAITutor)
  layouts/          # AppShell, AuthLayout, RoleLayout
  router/           # Route definitions, guards, lazy imports
  store/            # Zustand stores (authStore, uiStore, offlineStore)
  styles/           # CSS variables, global reset, typography
  utils/            # Formatters, validators, constants
```

10.2 Authentication Architecture

Concern	Implementation	Detail
Token storage	Access token: memory (Zustand store). Refresh token: httpOnly cookie (set by backend).	XSS cannot read httpOnly cookies. Access token in memory clears on page unload.
Silent refresh	Axios request interceptor calls /auth/token/refresh/ on 401.	Learner session continues without forced logout every 5 minutes.
Route guards	ProtectedRoute component wraps all authenticated routes.	Checks authStore.isAuthenticated. Redirects to /login if false. Stores intended URL for post-login redirect.
Role routing	RoleRouter component reads authStore.role.	Learner → /learner/ routes. Teacher → /teacher/ routes. MoES → /moes/ routes. Shared routes accessible to all.
Logout	Calls /auth/logout/ API, clears Zustand store, invalidates React Query cache, navigates to /login.	Ensures all in-memory state is wiped.
Persistent login	On app load: attempt silent token refresh. If succeeds, restore session. If fails, show login.	No stale JWT in localStorage.

10.3 State Management Strategy

State Type	Tool	Examples
Auth state	Zustand — authStore	user, role, isAuthenticated, accessToken
UI/modal state	Zustand — uiStore	sidebarOpen, activeModal, aiPanelOpen, offlineBanner
Offline state	Zustand — offlineStore + Workbox	isOffline, cachedLessons[], lastSynced
Server data	React Query	All API responses: lessons, posts, progress, sessions
AI panel context	Zustand — aiStore	currentContext (lesson/post payload), sessionId, messages[]
Form state	React Hook Form (local)	Registration, Login, CreatePost, UploadMaterial
WebSocket state	Custom useWebSocket hook	CollabSession messages, notification events

10.4 React Query Patterns

All API data is fetched, cached, and invalidated through React Query. The following query key conventions apply:

- ['lessons', subjectId, levelId] — lesson list queries
- ['lesson', lessonId] — single lesson detail
- ['posts', filters] — knowledge wall feed
- ['progress', learnerId] — competency progress
- ['ai-sessions', learnerId] — AI session history
- ['analytics', filters] — MoES dashboard data

Mutations (POST/PATCH/DELETE) always invalidate the relevant query key on success, triggering background refetch. Optimistic updates are applied for reactions (LIKE/SHARE) and post creation to give instant UI feedback before the server confirms.

10.5 AI Tutor Component Architecture

The AI Tutor exists as three nested component layers:

- AITutorPage — full-screen dedicated page (route: /ai-tutor). Contains SessionSidebar + ChatInterface.
- AIPanel — right-panel / bottom-sheet inline version. Triggered from content cards. Shares ChatInterface component.
- AIButton + AIDropdown — the contextual button on every content card. Renders a Floating Menu with suggested questions. On selection, populates aiStore.currentContext with the content payload and question, then opens AIPanel.

The aiStore holds the active context object: { contentType, contentId, contentTitle, contentBody, subjectId, competencyIds, question }. This object is included in every API call to /api/v1/ai/query/ so the backend AI layer receives full context for prompt enrichment via MCP.

10.6 Routing Structure

```
/ → redirects based on auth state
```

```
/login → S-AUTH-02
/register → S-AUTH-01
/learner/home → S-LEARN-01 (ProtectedRoute, role: LEARNER)
/learner/subjects → S-LEARN-02
/learner/lessons/:lessonId → S-LEARN-03
/learner/offline → S-LEARN-04
/learner/progress → S-LEARN-05
/learner/posts → S-POST-01
/learner/posts/new → S-POST-02
/learner/posts/:postId → S-POST-03
/learner/sessions → S-COLLAB-01
/ai-tutor → S-AI-01 (accessible to all authenticated roles)
/teacher/home → S-TEACH-01 (ProtectedRoute, role: TEACHER)
/teacher/materials → S-TEACH-03
/teacher/materials/upload → S-TEACH-02
/moes/dashboard → S-MOES-01 (ProtectedRoute, role: ADMIN)
/moes/curriculum → S-MOES-02
/moes/reports → S-MOES-03
/notifications → S-SHARED-01
/profile → S-SHARED-02
```

Part 11 — Optimised Meta-Prompt for Claude

The following is the precision-optimised version of the original design brief for use in Claude and other AI design generation tools. Terms that do not apply to UI/UX design have been replaced with accurate equivalents.

OPTIMISED PROMPT — CBC Platform UI Design Brief (Claude / AI Tool Optimised)

Design the CBC Digital Learning Platform — Uganda's national secondary school edtech web application (S1–S6 learners, teachers, MoES/NCDC officials) — as a fully responsive React SPA decoupled from a Django/DRF backend via REST API and WebSocket. DESIGN SYSTEM Palette: Deep Blue #1B4F72 (primary), Uganda Green #1E8449 (secondary), Warm Amber #D4A017 (CTA/accent), Deep Teal #0E6655 (AI Tutor brand). Typography: DM Serif Display (headings), Plus Jakarta Sans (UI/body), JetBrains Mono (AI output, code). 8px base grid, 12-column layout, max-width 1200px. LAYOUT PATTERNS Shell: dark top navigation bar + collapsible left sidebar (240px) + fluid main content + optional contextual right panel (320px). Mobile: top bar + bottom navigation (5 tabs) + bottom sheet (replaces right panel). All layouts WCAG 2.1 AA compliant, minimum 44px touch targets. KEY SCREENS Authentication (role-aware login, 2-step registration), Learner Home (personalised feed, subject browser, lesson cards), Lesson Reader (reading-optimised, competency progress, offline badge), Knowledge Wall (peer posts with inline comments and reactions), Progress Dashboard (competency attainment by subject, animated progress rings), Teacher Material Manager (upload form, CBC tagging, sharing toggle), MoES Analytics Dashboard (national engagement metrics, regional map, competency charts). CONTEXTUAL AI TUTOR INTEGRATION (defining UX pattern) Every content card (lesson, post, teacher material) carries a 32px teal circular AI button. On hover/tap: a smart dropdown renders 3 contextually generated suggested questions + a custom question input. On submit: the content payload + question are sent to the AI API. The AI response appears in a right panel (desktop) or bottom sheet (mobile) without page navigation. Additionally: a full-screen AI Tutor page (dedicated route /ai-tutor) with session history sidebar and persistent chat interface — AI always aware of CBC curriculum context via MCP knowledge base. FRONTEND ARCHITECTURE State: Zustand (auth, UI, AI context, offline) + React Query (all server data, optimistic updates). Auth: JWT in-memory access token + httpOnly refresh token cookie, Axios interceptor for silent refresh. Routing: React Router v6 with role-based ProtectedRoute guards and lazy loading. Offline: Vite PWA plugin (Workbox service worker) + IndexedDB for downloaded lesson bundles. Forms: React Hook Form + Zod. WebSocket: custom hook for real-time collaboration and notifications. DESIGN PRINCIPLES Consistent 8px spatial rhythm. Component-driven architecture. Progressive disclosure (show more on demand). Graceful degradation for offline and AI-unavailable states. Mobile-first responsive breakpoints. Optimistic UI for social interactions (reactions, comments). Clear visual hierarchy distinguishing CBC content (blue/green) from AI context (teal) from peer content (amber accents).

Corrections to Original Brief

The following terms from the original brief were not applicable to UI/UX design and have been substituted with the correct equivalents:

Original Term	Category	Correct UI/UX Equivalent
"State handling"	Inaccurate UI term	State management (Zustand, React Query, local component state)
"Logic"	Backend/generic term	Component behaviour, interaction logic, conditional rendering, event handlers
"Authentication logic"	Too broad	Auth flow: JWT storage strategy, silent token refresh, route guards, role-based navigation

Original Term	Category	Correct UI/UX Equivalent
"Locative UIs"	Non-standard term	Context-aware / adaptive UI — interfaces that respond to user context (role, location, connectivity)
"Embed AI models into UI"	Technically imprecise	AI-integrated UI patterns — embedding AI interaction surfaces (panels, buttons, dropdowns) that call AI APIs
"Design patterns for AI models"	Ambiguous	Conversational UI patterns, contextual AI affordances, AI response component design

Document Sign-off

This Stage G UI/UX Design Prompt Compendium establishes the complete design specification for the CBC Digital Learning Platform frontend. All Google Stitch prompt outputs, all component implementations, and all responsive behaviour shall conform to the design system defined in Part 1 and the architectural specifications in Part 10.

Role	Name	Signature / Date
Project Supervisor	_____	_____
UI/UX Designer / Group Leader	_____	_____
Frontend Architect	_____	_____